

Frontal Theta Phase Synchrony and Post-Error Slowing: A Psychobiological and Psychometric Investigation of Error Monitoring in a Large Open EEG Dataset

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Proposal Declaration

I, Moyosore Olasupo, declare that this proposal has been composed solely by me and has not been submitted, in whole or in part, in any previous application for a degree. Except where stated otherwise, by reference or acknowledgement, the work presented is entirely my own and has been generated by me as a result of my own original research. I acknowledge using Open AI's GPT-6 Astra to support my understanding of concepts, provide feedback, and assist with planning. I have not plagiarised AI generated content and I have reviewed the materials included in this proposal to ensure that it reflects my own understanding.

Signed:

A handwritten signature in black ink, appearing to read 'Moyosore Olasupo', written over a horizontal line.

Date: 13/09/2026

Background Literature Review

Successful goal-directed behaviour requires individuals to monitor their actions and adjust their responses when a mistake occurs. However, detecting an error does not necessarily improve subsequent performance (Danielmeier & Ullsperger, 2011). The study of how neural error signals are related to behavioural adjustment is important for understanding cognitive control. My literature review search was conducted by using APA PsycINFO, covering error monitoring, frontal theta activity, post-error slowing (PES), and reliability measurement. Error-related negativity (ERN) is an event-related potential that occurs shortly after an incorrect response and is measured at frontocentral scalp electrodes, including during flanker tasks (Holroyd & Coles, 2002). It provides an established measure of error processing. Theta power describes the strength of activity within a frequency band and is related to squared amplitude. Inter-site phase synchrony describes the consistency of phase relationships between signals recorded at different electrodes. These measures capture different aspects of the neural response to errors.

Prior Findings

Cavanagh et al. (2009) found increased theta synchrony between midfrontal and lateral-frontal EEG sites during error trials in a flanker task. Greater synchrony was associated with subsequent post-error slowing (PES), suggesting a relationship between synchronised frontal activity and behavioural adjustment. Valadez and Simons (2018) additionally found that error-related frontal theta power and PES predicted performance recovery. However, findings

concerning theta power do not directly establish the behavioural relevance of connectivity, and these associations alone do not demonstrate causation.

Danielmeier and Ullsperger (2011) reviewed evidence that PES and improved post-error accuracy are not consistently associated. This means that a slower response does not automatically indicate successful adaptation. Murphy et al. (2016) found that larger error-evoked pupil responses predicted greater slowing and a higher probability of a correct subsequent response in a task with long inter-stimulus intervals. This suggests that the timing of subsequent task demands influences whether an error response supports beneficial adjustment. Volume conduction poses a challenge for interpreting scalp connectivity. Vinck et al. (2011) introduced the weighted phase-lag index (wPLI) to estimate phase synchrony with reduced sensitivity to volume conduction effects and noise. They also developed a debiased estimator of its squared value to reduce sample-size bias. This provides a basis for estimating lagged connectivity.

Reliability must be assessed when interpreting individual differences. Clayson et al. (2021) distinguished measurement precision from internal consistency, explaining why good data quality does not automatically establish reliable measurement of differences between individuals. Consequently, the reliability of error-related theta synchrony should be assessed alongside its relation with behaviour.

Theory

The reinforcement-learning account proposes that the ERN reflects a negative learning signal carried through the dopamine system to the anterior cingulate cortex, contributing to subsequent performance adjustment (Holroyd & Coles, 2002). This provides a connection between error monitoring and behavioural adaptation. A complementary network interpretation proposes that theta synchrony supports coordination between frontal regions involved in performance monitoring and implementing adjustments. The findings of Cavanagh et al. (2009)

are consistent with this interpretation, but do not establish direct communication between particular anatomical regions.

A positive association between theta connectivity and PES would support a neural–behavioural relationship. Examining this relationship separately from measurement reliability allows the study to address both the behavioural relevance and psychometric properties of frontal theta connectivity.

Rationale and Hypotheses

Previous findings provide a basis for examining whether error-related frontal theta synchrony is associated with PES (Cavanagh et al., 2009). This study will investigate the relationship between participants using the ds004883 flanker dataset. A secondary comparison will test whether connectivity is more reliable than ΔERN ; this advantage has not yet been established. This study will use the publicly available EEG dataset ds004883. Its retained behavioural information supports trial reconstruction, although final eligibility for the planned analyses will depend on the specified screening criteria.

The research questions are: **Is error-related frontal theta phase synchrony associated with post-error slowing (PES) between participants? Does error-trial theta synchrony demonstrate higher internal consistency than ΔERN at matched numbers of usable error trials?**

H1: Error responses will produce a more negative mean response-locked ERP amplitude at FCz than correct responses during 0–100 ms after the response. Prediction: $\Delta\text{ERN} = \text{ERN} - \text{CRN} < 0$.

H2: Frontal theta synchrony, measured using dWPLI between FCz and lateral-frontal electrodes, will be greater on error trials than correct trials.

H3: Individuals with stronger error-related theta synchrony will show greater PES. $PES = RT(E+1) - RT(E-1)$, using valid correct responses immediately surrounding an error.

H4a: Error-trial theta dWPLI will show higher Spearman–Brown-corrected split-half reliability than ΔERN at matched numbers of usable error trials.

H4b: Examine the stability of the H4a reliability comparison across repeated random splits of the same trials. This is a robustness analysis, not a separate formal hypothesis.

Aims and Objectives

The primary aim is to examine the relationship between error-related frontal theta synchrony and PES, and the secondary aim is to compare connectivity and ΔERN reliability. The objectives are to (i) evaluate error-related EEG effects, (ii) test the between-participant association between synchrony and PES, and (iii) compare reliability using matched error-trial counts.

Research Design

Design

The study will use a quantitative secondary analysis of EEG and behavioural data from the ds004883 flanker dataset (Clayson et al., 2024). It will combine within-participant comparisons of error and correct responses with a between-participant analysis of frontal theta synchrony and post-error slowing (PES). No additional participants will be recruited.

H1 will compare mean response-locked ERP amplitudes between error and correct trials at FCz.

H2 will compare theta synchrony between the response conditions.

H3 will examine whether participants with stronger error-related connectivity have higher PES.

H4a will compare the Spearman–Brown-corrected split-half reliability of connectivity and Δ ERN using matched numbers of usable error trials. H4b will assess stability across repeated random splits of the same trials.

Materials

The study will use EEG recordings and associated behavioural metadata from the ds004883 flanker dataset (Clayson et al., 2024). The original study presented three arrow-flanker task variants using E-Prime 3.0. EEG was recorded using an Electrical Geodesics, Inc. HydroCel Geodesic Sensor Net with 128 scalp sites and a vertex reference, sampled at 500 Hz. The dataset contains EEGLAB SET/FDT files, electrode information and event metadata. Behavioural fields retained in the SET files include response times, response accuracy and original trial identifiers. These fields will provide information for reconstructing behavioural sequences.

EEG processing and analysis will use MNE-Python 1.12.1, supported by NumPy, SciPy, and Matplotlib. Picard will be used for independent component analysis. The standard montage provides approximate correspondences between E6 and FCz, E27 and F5, and E123 and F6. These represent standard electrode locations rather than individual anatomical localisation.

Participants

The source dataset contains records for 172 participants recruited from undergraduate courses at Brigham Young University and the University of South Florida. According to Clayson et al. (2024), participants received course credit, and the eligibility criterion specified an age over 18 years. Eligibility excluded individuals reporting psychiatric or neurological diagnoses, psychoactive medication use, uncorrected visual impairment or limited English fluency.

All participants whose data meet the prespecified, analysis-specific eligibility criteria will be considered for the secondary analysis. Inclusion will depend on EEG quality and sufficient usable error and correct trials. The synchrony PES analysis will additionally require sufficient valid behavioural sequences. Consequently, eligible sample sizes may differ across the ERP, connectivity, behavioural-association and reliability analyses.

Participants will be included according to predefined data-quality and trial-count criteria. The final sample size and reasons for exclusion will be reported for each analysis. Statistical sensitivity will be evaluated using the number of eligible participants.

Data Analysis Plan

Before hypothesis testing, response-time exclusions, minimum usable trial counts, task-variant handling, connectivity parameters and statistical decision rules will be specified. EEG preprocessing will include filtering, referencing, ocular-artefact correction and response-locked epoching, with documented quality checks and exclusions. Original trial

identifiers will be preserved when linking EEG epochs to behavioural sequences. Recordings from the same participant will not be treated as independent observations.

For H1, ERN and CRN will be calculated as the mean response-locked ERP amplitude at FCz/E6 during 0–100 ms following error and correct responses, respectively. The difference score will be calculated as $\Delta\text{ERN} = \text{ERN} - \text{CRN}$, and the response-locked ERP waveforms will be plotted. Error and correct amplitudes will be compared within participants using an appropriate prespecified group-level test.

For H2, frontal theta synchrony will be estimated using a debiased weighted phase-lag estimator between E6 and the approximate lateral-frontal sites E27 and E123. The proposed frequency band is 4–8 Hz, with a 0–150 ms post-response scoring window. The exact estimator, electrode-pair aggregation, wavelet parameters and sufficient signal duration will be specified before testing. Error- and correct-trial estimates will be compared within participants using a predefined approach to trial-count matching. The implications of an unsupported H2 result for interpreting H3 will also be specified in advance.

For H3, post-error slowing will be calculated as $\text{PES} = \text{RT}(\text{E}+1) - \text{RT}(\text{E}-1)$, using valid correct responses immediately before and after an error (Dutilh et al., 2012). Triplets will be screened for omissions, incorrect flanking responses, response-time exclusions and crossings of task-block boundaries or recording discontinuities. Neighbours will be identified from the original trial sequence before exclusions are applied. The association between participant-level error-trial synchrony and mean valid PES will be assessed using Spearman's correlation.

For H4a, Spearman–Brown-corrected split-half reliability will be compared between error-trial synchrony and ΔERN using matched usable error trials and the same eligible participants within each comparison. Correct-trial sampling for ΔERN will be predefined. Reliability will be examined across a prespecified range of usable error-trial counts, informed by

dataset feasibility and relevant methodological evidence. Each comparison will use matched trial counts for dWPLI and Δ ERN, with the number of trials contributing to each half stated explicitly.

For H4b, repeated random split-half partitions will assess how sensitive the reliability comparison is to the division of trials. Results will be summarised descriptively. Because the partitions reuse the same participants and trials, their reliability coefficients will not be treated as independent observations.

Effect sizes, uncertainty intervals, usable trial counts, participant exclusions and analysis-specific sample sizes will be reported. A multiple-testing strategy will be specified before confirmatory testing, and subsequent departures from the agreed plan will be identified as exploratory.

Procedure

The study will use existing EEG and behavioural data from the ds004883 dataset. Trial-level behavioural information will be assessed to establish whether reaction times, response accuracy and trial sequence support the calculation of a meaningful post-error behavioural outcome.

EEG and behavioural measures will be derived using the specified analysis procedures. The primary analysis will examine the relationship between error-related frontal theta synchrony and post-error slowing. A secondary analysis will compare the Spearman–Brown-corrected split-half reliability of theta synchrony and Δ ERN using matched numbers of usable error trials. Repeated random splits will assess the stability of this comparison without treating the resulting

coefficients as independent observations. Analysis-specific exclusions, usable trial counts and participant numbers will be documented.

Ethics

The study will use existing de-identified EEG and behavioural data from the publicly available ds004883 dataset. Clayson et al. (2024) report that participants provided written informed consent covering public data sharing and its associated risks. No participants will be contacted and no new data will be collected.

The project will follow the University of Essex Online's Research Ethical Approval Policy and programme-specific guidance (University of Essex Online, 2025). The required application documents and review arrangements will be confirmed with the module tutor.

No attempt will be made to identify participants or link their records to identifying information. Original data will remain unchanged, with derived outputs stored separately. Storage, access and backup arrangements will be documented once confirmed and checked against university requirements.

Limitations

This study uses an existing dataset, so the available task design, recordings and behavioural information cannot be modified. Differences between the three flanker variants may influence EEG responses and post-error behaviour, requiring explicit consideration in the analysis.

Participant eligibility will depend on EEG quality, usable error trials and valid behavioural sequences. Exclusions may reduce the analysable sample and limit generalisability. Small error-trial counts may also produce imprecise synchrony and reliability estimates, particularly when trials are divided into halves.

Post-error slowing is a behavioural index rather than a direct measure of adaptive cognitive control. Comparing responses immediately before and after errors addresses some local fluctuations in response speed but does not eliminate all influences, including trial difficulty and congruency. Uncertainty about the correspondence between recorded event markers and physical stimulus timing also limits interpretation of absolute reaction times.

Scalp EEG provides limited anatomical specificity. Although dWPLI reduces sensitivity to some zero-lag effects and sample size bias, it does not eliminate artefact contamination or establish direct communication between brain regions. Standard electrode correspondences do not represent individual anatomical localisation, and time frequency smoothing limits the temporal specificity of connectivity estimates.

Finally, associations between synchrony and PES cannot establish causation. Higher split-half reliability would indicate greater internal consistency under the chosen conditions, but would not by itself establish greater validity or stability across separate recording sessions.

Timeline

Table 1

Proposed Research Timeline

Work package	Main activities	Indicative timing
1. Finalise research protocol	Agree hypotheses, screening criteria and analysis settings.	Week 1
2. Ethics review	Submit the ethics application and address any requested revisions.	Weeks 2–3
3. Behavioural screening	Apply agreed exclusions to existing trial records	Weeks 4-5
4. EEG preprocessing	Extend the reviewed pipeline and record usable trials	Weeks 5–7
5. Calculate measures	Calculate PES, Δ ERN, and theta synchrony	Weeks 7–9
6. Statistical Analysis	Test H1-H3 and conduct H4a reliability and H4b robustness analyses	Weeks 10–12
7. Write Up	Complete results, discussions and limitations	Weeks 13–14
8. Revision and submission	Proofread, check formatting and references	Week 15

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